

Sonok Mahapatra

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EDUCATION

University of Maryland, College Park

Expected Dec 2027

B.S. Computer Science & Applied Math; Minor in Comp. Finance

GPA: 3.7

Relevant Coursework: Grad High Performance Computing, Grad Signal Processing, Grad Stochastic Processes, Advanced Statistics, Machine Learning, Computer Systems, Advanced Algorithms, Equations, Abstract Algebra, Data Science, Object Oriented Programming

AWARDS & CERTIFICATIONS

Putnam Top 700 Scorer (14/120); Five-time AIME Qualifier ('18-'22); Dean's List; 2nd Place Lockheed Martin CodeQuest; Science Bowl Quarterfinalist; Presidential Scholar; [Bloomberg Financial Fundamentals Certification](#); [AWS Certified Cloud Practitioner](#)

EXPERIENCE

Amazon Web Services

May 2026 – Aug 2026

Software Development Engineer Intern, Networking Infrastructure

- Incoming SDE intern on the AWS Networking Infrastructure team; will work on low-latency distributed systems and high-throughput data plane software at scale.
- Developing scalable data ingestion and processing pipelines for high-volume network telemetry, enabling efficient transformation and analysis of streaming data.
- Leveraging distributed systems and stream processing techniques to handle large-scale data flows, with a focus on latency, fault tolerance, and pipeline reliability.

Probanker

May 2025 – August 2025

Software Engineering Intern

- Scaled [ProBanker's](#) Django multi-tenant backend for 1K+ users by hardening async job execution (Celery) and optimizing SQL-heavy ORM workflows for simulations and reporting.
- Designed and optimized ETL pipelines for interbank market simulations (batch ingestion, feature generation, and rollups via Airflow-based scheduling), cutting end-to-end runtime by **40%** through vectorization and C/C++ extensions.
- Parallelized the simulation core using OpenMP and cache-aware memory layouts; built monitoring dashboards to track pipeline health and throughput under large scenario runs.

RESEARCH

State Street

May 2023 – Aug 2023

Lead Project Developer

- Co-led an equity sentiment forecasting initiative with State Street, generating alpha signals from SEC filings, macroeconomic factors, and market microstructure data.
- Built scalable ETL pipelines on Google Cloud to ingest, transform, and load 10K+ daily SEC filings and financial news into analysis-ready feature stores using SQL and Python.
- Developed and backtested BERT and VaderSentiment-based NLP factor models, reducing out-of-sample MSE by 20% and improving predictive IR; surfaced results via interactive Tableau dashboards.
- Published a detailed [project overview](#) and open-sourced the [frontend code](#) to support reproducibility and cross-team research.

University of Maryland

Jun 2024 – Aug 2024

Data Engineering & Analysis – AWS Network Traffic Patterns

- Built an end-to-end ETL pipeline to ingest, clean, and feature-engineer a 450K+ record cybersecurity dataset from an AWS-hosted honeypot; wrote complex SQL queries for aggregation and exploratory analysis.
- Engineered features from raw network logs (country mapping, port activity rollups, protocol encoding, time-of-day bucketing) using Pandas and PySpark; resolved schema inconsistencies at scale.
- Trained a Random Forest classifier for anomaly detection; evaluated with precision, recall, and F1-score and used feature importance to identify highest-signal attributes.
- Built interactive dashboards and traffic heatmaps using Matplotlib and Seaborn; automated the analysis pipeline end-to-end with CI/CD via Git and Docker.

TECHNICAL PROFICIENCIES

Languages: Python (NumPy, Pandas, StatsModels), C/C++, Rust, SQL, MATLAB, OCaml, JavaScript/TypeScript, Java

Data & ML: Spark, Hadoop (Hive/MapReduce), Airflow, PyTorch, Scikit-learn, XGBoost, Tableau, TensorFlow

Infrastructure: Django, FastAPI, Flask, React, Docker, Kubernetes, Jenkins, Git, Google Cloud Platform, PostgreSQL, Redis